

Tech Shipping Operations Automation — Final Master Plan

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Vision

Tech Shipping will run every order, from a master's WhatsApp message to delivery on board, through one AI-assisted platform. The goal: a quotation ready for approval within 30 minutes of an inquiry arriving, instead of hours or days.

The platform reads supply lists automatically, matches items to known vendors and past prices, calculates margin, GST and port charges, and tracks negotiation, purchase, delivery and payment. Staff approve every key step; the software does the typing, searching and arithmetic.

Build approach: one developer, working with AI coding tools (Claude Code), on a deliberately simple stack. Quotations go live in about 2 weeks; the full system is in use in about **10–12 weeks**, then improves continuously. The detailed build pack for Claude Code is in four files: `CLAUDE.md`, `SPEC.md`, `BUILD_PROMPTS.md` and `.env.example`.

Working name: ChandleryOS (to be renamed).

Design principles: keep it simple

Every choice below favours fewer moving parts, so one person can build, deploy and maintain the system.

1. **One app, one database.** A single Django application and a single PostgreSQL database. No separate frontend app, no extra vector database, no queue server.
2. **Use what comes free.** Django's built-in admin gives logins, user roles and data-entry screens for vendors, items and ports on day one.
3. **Money is code, not AI.** Commission, GST and port charges are fixed, tested calculations. AI never decides a price.
4. **AI reads, searches and drafts; people approve.** Nothing reaches a client or vendor without a person clicking approve.
5. **Deploy by pushing code.** The app runs on a managed platform that redeploys automatically from GitHub and backs up the database daily.

6. **Ship something useful every two weeks.** Each sprint ends with a working feature the team uses on real inquiries.
7. **Add AI agents one at a time**, only after the foundation works and each agent has proven reliable.

Business process: today vs. automated

Every step below is done by hand today. The software automates the typing, lookups and calculations; people keep the decisions.

Step	Today (manual)	With the software
1. Inquiry arrives	Master or vessel company sends list on WhatsApp (mostly) or email	Captured automatically in one inbox
2. Understand items	Staff read messy lists and photos	AI extracts item, spec, quantity, unit; staff confirm
3. Find vendors	Internet search and phone calls every time	Vendor database suggests past vendors and prices; AI helps find new ones
4. Collect vendor prices	Calls, WhatsApp, email	RFQs sent from the system; replies captured
5. Build quotation	Manual math in Excel	Automatic: landed cost + margin (20–40%) + port charges + GST
6. Send quotation	Typed and sent by hand	Branded PDF sent on WhatsApp or email in one click
7. Negotiation	Scattered chats, no record	Every revision saved; floor-price warning
8. Buy from vendor	Manual orders	Purchase orders generated automatically
9. Delivery	Tempo (local) or train (distant ports), tracked by phone	Dispatch record, status updates, proof of delivery
10. Invoice and payment	Separate billing	GST invoice from final quote; payment tracking

Step	Today (manual)	With the software
11. Learning	Knowledge stays in people's heads	Every price and vendor result saved and reused

Real-world cases the system handles

Situation	How the system handles it
Client adds items later ("add 5 more ropes")	The message is attached to the same open inquiry; new lines are added to it
Item never sold before	Staff create it from the inquiry line; a Manager sets HSN and GST before the quote can be approved
Item not available	Line marked "Not available" or offered as a substitute on the quotation
Client accepts only some items	Each line is accepted or not; orders, deliveries and invoices use accepted lines only
Negotiation on one item	Manager overrides that item's selling price; margin is recalculated and the floor warning still applies
Client and vendor units differ	Conversion factor, e.g. vendor sells a 25 KG drum, client orders in KG
Advance payment before buying	Proforma invoice; advance recorded and later applied to the tax invoice
Partial delivery, shortage or return	Deliveries record quantities per line; returns after invoicing use a GST credit note
Paying vendors	Vendor bills and payments recorded against purchase orders, so real profit per order is known
Lists in Hindi, Marathi or mixed language	AI reads them and writes item names in English, keeping the original wording

The WhatsApp problem and the fix

Recommendation: convert the co-founder's existing number to the WhatsApp Business app, then connect it to the software using Meta's **Coexistence** feature. Clients keep messaging the same number, and the co-founder keeps using it on his phone.

The problem. Clients currently message the co-founder's personal WhatsApp. Software cannot legally connect to a personal WhatsApp account, so inquiries cannot be captured automatically. Business and personal chats are also mixed on one phone.

Why Coexistence solves it. Meta now lets one WhatsApp Business app number work on the phone and through the official **Cloud API** (the software connection) at the same time. Existing chats stay, no number change is needed, and new messages sync both ways in real time. [source](#) · [Meta docs](#)

Migration steps:

1. Back up the co-founder's WhatsApp chats (Google Drive or iCloud backup).
2. Install the WhatsApp Business app on his phone and register the same number.
Personal WhatsApp chat history transfers across.
3. Set up the business profile: Tech Shipping name, logo, address, hours, email.
4. Create a Meta Business portfolio and verify Tech Shipping (GST certificate, company documents).
5. Onboard the number to the Cloud API through the Coexistence flow (a QR code scanned from the Business app).
6. The software then receives every inquiry automatically; staff can still reply from the phone.

Things to know:

- The number becomes a business account. The co-founder should move purely personal chats to a separate personal number over time.
- If the phone is changed or WhatsApp reinstalled, the software link drops and reconnects after re-registration.
- Old chats (before the switch) are not pulled into the software automatically. Past supply lists can be exported with WhatsApp's "Export chat" and loaded as history.
- Avoid unofficial WhatsApp automation tools (WhatsApp Web bots). They break WhatsApp's terms and risk the number being banned.
- Meta's API message pricing changes from October 1, 2026; check current rates before launch. [source](#)

Alternative: a new dedicated business number on the Cloud API, announced to all clients. Cleaner separation, but clients must switch numbers.

WhatsApp bot

The WhatsApp Cloud API is the bot API. Through Coexistence, the co-founder's number gets it without changing numbers: he keeps chatting on his phone while the software and bot use the same number.

What the bot does:

- Acknowledges each inquiry instantly and confirms when the quotation will follow
- Reads supply lists (text, photo, PDF, Excel) and passes them to the software
- Asks for missing details: vessel name, port, ETA, delivery date
- Sends the quotation PDF after staff approve it
- Sends status updates: order confirmed, dispatched, delivered on board
- Sends payment reminders for pending invoices
- Hands over to a person when negotiation starts or it is unsure

Meta's rules:

- **24-hour window:** after a client messages, the bot can reply freely for 24 hours; after that, only pre-approved template messages.
- **Templates:** dispatch updates, payment reminders and "quote ready" messages are written once and approved by Meta.
- **Charges:** some message types are charged per message; rates change from October 1, 2026, so confirm before launch.
- **Verification:** Tech Shipping must be verified on Meta Business (GST certificate, company documents).

Limitations:

- The bot cannot read WhatsApp group chats; masters who send lists in groups should message the number directly.
- The co-founder and the bot see the same chats. Rule: the bot acknowledges, collects details and sends documents; negotiation stays human.
- Some app features (such as disappearing messages and broadcast lists) may not work on a Coexistence number; confirm the exact list during setup.
- Only new messages reach the software; old chat history stays on the phone.

How to connect:

Option	Pros	Cons
Direct Meta Cloud API (recommended)	No middleman fee, full control, fits custom software	Needs developer setup
Provider (Interakt, AiSensy, WATI, Gupshup, etc.)	Faster setup, ready shared inbox	Monthly fee on top of Meta charges; less control

A provider is worth it only if WhatsApp automation is needed before the platform is ready.

Software modules

The platform has ten modules. Each row shows what it does and the technical terms involved.

#	Module	What it does	Technical terms
1	Inquiry intake	Reads WhatsApp and email inquiries, including photos, PDFs and Excel lists; extracts each line item	WhatsApp Cloud API, IMAP/SMTP email, Claude vision (reads photos, no separate OCR), structured extraction
2	Item master	One standard name per product with IMPA code, HSN code, category, unit and GST rate	Master data, normalisation, embeddings stored with pgvector
3	Vendor management	Vendors worldwide, products, contacts, lead times, full price history, scorecard; AI search for new vendors	RFQ, vendor scoring, AI agent, web search
4	Quotation engine	Calculates landed cost, margin, port charges and GST; produces branded PDF quote	Pricing engine, business rules, PDF generation

#	Module	What it does	Technical terms
5	Negotiation tracker	Saves every revision, discount and reason; warns below minimum margin	Version history, audit trail, floor-price rule
6	Purchase and delivery	Purchase orders from accepted lines; dispatch by tempo, train or courier; partial deliveries; proof of delivery	Workflow, state machine, PO, POD
7	Port charges library	Port-wise charges (Mumbai, JNPT, Hazira, Kandla and others) applied automatically	Reference data, rules table
8	Invoicing and accounts	Proforma and GST invoices, advances, credit notes, vendor bills and payments, real profit per order, Excel export for the accountant	Proforma invoice, credit note, payables (e-invoicing and Tally later)
9	Dashboard and reports	Open inquiries, pending quotes, deliveries, revenue, margin, conversion rate	KPIs, BI dashboard
10	AI copilot	Staff ask questions in plain English or Hindi and get answers from company records	RAG, pgvector, tool calling, prompt engineering

Every AI step is **human-in-the-loop**: staff confirm extracted lists and approve quotes before anything reaches a client or vendor.

AI agents

AI is used only where reading messy human input is the hard part. Everything else is plain, predictable code. After reviewing each candidate agent, most turned out to work better as simple rules.

Task	Approach	Why
Read supply lists (text, photo, PDF, Excel)	AI (Claude with vision), saved as drafts staff confirm	Lists are messy and handwritten; rules can't read them
Read vendor quotations	AI , saved as draft prices staff confirm	Every vendor formats differently
Spot follow-up messages	AI suggests , staff attach	Wording varies; the final decision stays human
Match item names	Plain code: remembered names, then fuzzy text search	Fast, free, explainable
Choose vendor per item	Plain rules: lowest recent landed cost, then lead time and reliability	Predictable; staff can override
Reminders and follow-ups	Approved WhatsApp templates, sent on click	No judgement needed
Vendor ratings	Simple calculations	No AI needed
Staff questions	Optional read-only assistant (Sprint 4)	Useful once data builds up
Finding new vendors on the web	Not built now; revisit after about 3 months	Staff do this well; adds cost and risk
Predictions	Not built now; revisit after 6+ months of data	Needs history first

AI accuracy is measured on 20+ real, anonymised lists with known correct answers; a prompt or model change is kept only if accuracy does not drop. If AI fails, the inquiry opens for manual entry, so work never stops.

Guardrails on every AI step:

- AI can create drafts but can never send a quotation, order or payment on their own.
- AI never calculates or changes prices, commission or GST.
- Every AI call is logged, so staff can see what it did and why.
- If the AI is unsure (unclear item, low confidence), it flags the item for a person instead of guessing.

How it is built: direct calls to the Claude API with a fixed output format, validated before saving. No agent framework is needed.

Pricing formula

GST is charged on the selling price, which already includes Tech Shipping's margin. The quotation engine calculates each line in this order:

1. **Vendor price** × exchange rate + **freight** = landed cost per vendor unit
2. ÷ **unit conversion** (e.g. 25 KG per drum) = **landed cost** per client unit
3. + **Margin** of 20–40%, or a Manager's price override = **selling price**
4. × quantity – **overall discount** = **taxable value**
5. + **GST** at the item's HSN rate: CGST + SGST if the port is in Maharashtra, IGST otherwise = **line total**

Not-available lines and lines the client did not accept are left out of totals.

Then for the whole quotation:

5. Sum of line totals
6. + **Port charges and delivery charges** (services, usually 18% GST)
7. = **Grand total**

$$\text{Line total} = \frac{\text{Vendor price}}{\text{Conversion}} \times (1 + \text{Margin}) \times \text{Qty} \times (1 - \text{Discount}) \times (1 + \text{GST rate})$$

The client's PDF shows selling price, GST rate and GST amount per line. Vendor cost and margin stay internal and are visible only to authorised staff.

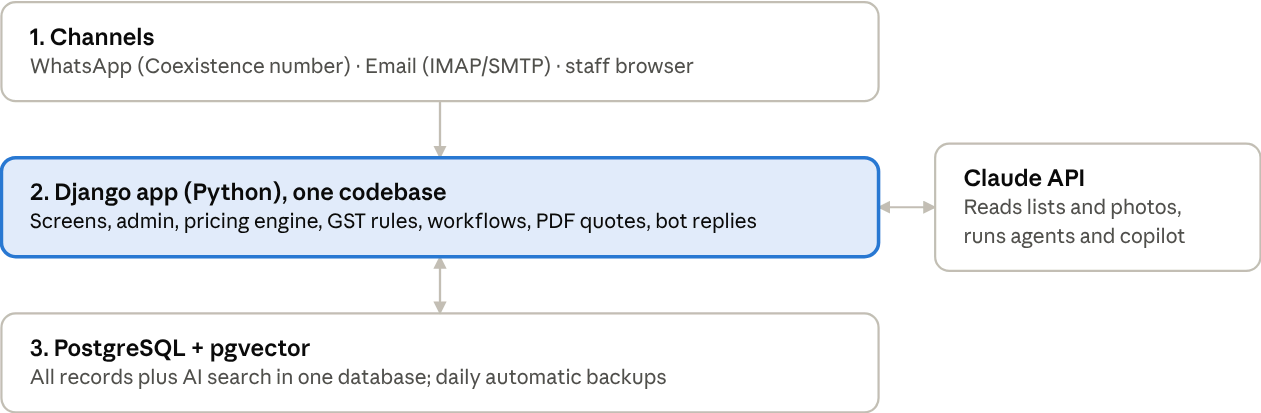
To confirm with the CA before building:

- Supplies to **foreign-going vessels** may qualify as exports and be zero-rated for GST, subject to conditions and documents. Supplies to **coastal vessels** are taxed normally. Each quotation will carry a vessel-type flag.
- Correct GST rate for each HSN code in the item master.
- Customs duty and IGST treatment if Tech Shipping imports goods directly.
- VAT handling for any supplies taxed outside India.

Architecture, tech stack and data model

The whole system is one Django app and one database, deployed by pushing code to GitHub.

One app and one database do all the work; AI helps from the side



system architecture · 5 layers + frontend

Messages arrive from WhatsApp and email into the Django app, which stores everything in PostgreSQL. The app calls the Claude API only to read lists, draft and answer questions; all pricing runs in the app's own tested code.

Technology stack

Part	Choice	Why it keeps things simple
Language	Python	One language for everything; best for AI
Web framework	Django	Built-in admin, logins, user roles and forms
Screens	Django templates + HTMX + Bootstrap 5	No separate frontend app, no build step; works on laptop and phone
Database	PostgreSQL with pg_trgm	One database; fuzzy item-name matching built in
AI	Claude API (vision + structured output)	Reads photos and PDFs directly

Part	Choice	Why it keeps things simple
Background tasks	Django-Q2 (runs on the database)	No Redis or queue server to manage
PDFs	WeasyPrint	Quotes, proforma, invoices, credit notes, POs from one template style
WhatsApp	Meta Cloud API, direct (Coexistence)	Official; keeps the existing number; no provider fee
Email	IMAP + SMTP	Works with the existing company mailbox, any provider
Files	Cloudflare R2 (private bucket)	Photos and PDFs shared safely by the app and background worker
Hosting	Render: production + staging	Auto-deploys from GitHub; managed database with daily backups
Checks	GitHub Actions	Tests run before every deployment
Error alerts	Sentry (free tier)	Sends an alert when something breaks

Removed from the earlier plan: separate React frontend, Qdrant, LangChain, Celery and Redis, separate OCR service, and hand-managed servers. Any of these can be added later if volume demands it.

If data must stay in India: host on a single AWS Lightsail server in Mumbai instead, with the same app. Slightly more setup, same code.

Safety in development: a staging copy uses a test WhatsApp number and fake demo data. Messages to real clients are blocked everywhere except production.

Main data tables

Area	Tables
Clients	Clients, Vessels (IMO, foreign-going or coastal)
Products	Categories, Items (IMPA, HSN, GST, pending review), Item aliases, Unit conversions

Area	Tables
Vendors	Vendors, Vendor prices (currency, exchange rate, freight, source)
Ports	Ports (state), Port charges
Inquiries	Inbound messages and media, Inquiries (with timing for the 30-minute goal), Inquiry lines
Quotes	Quotations and revisions, Quote lines (availability, conversion, override, accepted), Quote charges
Orders	Purchase orders and lines, Deliveries and delivery lines
Billing	Proforma invoices, Tax invoices, Payments and allocations, Credit notes, Vendor bills, Vendor payments
System	Company settings, Users and roles, Notifications, AI call log

Build phases

Built in four sprints by one developer using Claude Code, with 19 ready-to-paste prompts in `BUILD_PROMPTS.md`. Each sprint ends with something the team uses on real inquiries. Company tasks in Week 0 run alongside Sprint 1, because Meta verification and data collection take time outside the code.

Stage	What gets built	When	Result for the business
Week 0 (in parallel)	WhatsApp Business switch, Meta verification, permanent token, templates; test number; 20 real lists with correct answers; vendors, port charges, GST rates; CA questions; accounts	Weeks 1–2	Approvals and data ready

Stage	What gets built	When	Result for the business
Sprint 1: Quotations	Setup, staging, tests on every push, master data and import, pricing engine, new items, quote builder, PDF, approval, revisions, notifications	Weeks 1–3	Quotations in minutes; first real quotes sent
Sprint 2: Automatic intake	WhatsApp and email, AI reading of lists and vendor quotes, follow-up messages, RFQs, AI accuracy tests	Weeks 4–6	No more typing lists or vendor prices
Sprint 3: Orders to cash	Partial acceptance, purchase orders, partial deliveries, proforma and advances, invoices, credit notes, vendor bills, real profit, vendor ratings, metrics	Weeks 7–9	Every order tracked from quote to payment
Sprint 4: Go-live	Security review, backup restore rehearsal, optional assistant, user and admin guides	Weeks 10–11	Safe, documented system
Parallel run	Team uses the system alongside Excel, then switches fully	Weeks 11–12	Confident switch-over
Later	Web vendor search, predictions, e-invoicing, Tally, warehouse stock	When needed	Smarter decisions from history

If time is short, cut first: the optional assistant, global search and advanced dashboard metrics. Sprints 1 and 2 alone deliver most of the value.

Team

Role	Responsibility
Developer (one person, with Claude Code)	Builds, deploys and maintains the whole system; reviews and tests all AI-written code
Project sponsor (co-founder)	Approves priorities, owns the WhatsApp number, signs off each sprint
Operations staff	Supply sample data, test with real inquiries, give feedback
CA / accountant (advisory)	Confirms GST, export and e-invoicing rules

Running costs (approximate, monthly)

Item	Approx. cost
Render production: web, background worker, managed database	Paid plans; check current Render pricing
Render staging (smallest plans)	Low; check current Render pricing
File storage (Cloudflare R2)	Very low at this volume
AI usage (Claude API)	About ₹2,000–10,000, depending on volume
WhatsApp Cloud API	Per-message charges set by Meta; check rates after Oct 1, 2026
Error alerts (Sentry), code and checks (GitHub)	Free tiers
Domain and email	About ₹500–1,000

These are rough estimates to be confirmed when choosing providers.

Risks

Risk	Mitigation
AI misreads an item or quantity	Staff confirm every extracted list before quoting
AI sends or changes something it shouldn't	AI can only draft; sending and pricing need a person; every action logged
Wrong GST rate or calculation	Rates tied to HSN codes, verified by CA, editable by admin only; pricing engine covered by automated tests
AI-written code nobody understands	Developer reviews every change, keeps code simple and documented, writes tests for money logic
Only one developer	Code on GitHub with setup notes; managed hosting; paid maintenance agreement
Messy existing data	Week 0 dedicated to collection and cleanup
Meta verification delays	Start in Week 0; Sprint 1 does not depend on WhatsApp
WhatsApp link drops on phone change	Documented reconnection steps; alert in software
Staff resistance	Quotations go live first; they save time from week two
Leak of margins or vendor prices	Role-based access, encrypted connections, daily backups
Personal data compliance	Follow the DPDP Act 2023; option to host in Mumbai

Success measures

- Inquiry to quotation sent: under 30 minutes
- Manual data entry: reduced by 80% or more
- Quote-to-order conversion rate: tracked monthly
- Average margin per order: visible and protected

- Zero calculation errors in quotations

Open decisions

- ☐ Migrate the co-founder's number (Coexistence) or start a new business number?
- ☐ Does Tech Shipping import directly (customs duty applies) or buy only from vendors delivering in India?
- ☐ Mostly foreign-going vessels, coastal, or both?
- ☐ How many staff will use it, and who can see margins and vendor costs?
- ☐ Laptop, phone, or both?
- ☐ Do some clients need quotes in USD?
- ☐ Share 5–10 real supply lists and 3–5 past quotations for Phase 1
- ☐ Hosting: Render/Railway (simplest) or a Mumbai server (data in India)?
- ☐ Is VAT ever charged (supplies at foreign ports)? Only Indian GST is built in now.
- ☐ Does Tech Shipping import directly? If yes, add a customs duty field to vendor prices.
- ☐ Invoice once per order, or per delivery?

Glossary

Term	Plain meaning
AI agent	An AI that carries out a task in steps: reads, searches, uses the system's functions, drafts a result
API	The way two software systems talk to each other
Audit trail	A record of who changed what, and when
Backend	The engine behind the screen: calculations and rules
Cloud API (WhatsApp)	Meta's official connection letting software send and receive WhatsApp messages
Coexistence	Meta feature letting one number work on the phone app and the Cloud API together

Term	Plain meaning
Django	A Python toolkit for building web apps, with logins and admin screens built in
Embeddings	Text turned into numbers so the computer can find items with similar meaning
Frontend	What users see and click
Guardrails	Rules that limit what an AI agent may do
HSN code	India's tax classification code; decides the GST rate
HTMX	A small tool that makes web pages update instantly without a separate frontend app
IMO number	Unique permanent ID of every ship
IMPA code	Marine industry's standard catalogue code for ship stores
LLM	Large Language Model: AI that understands and writes text (e.g. Claude)
MVP	Minimum Viable Product: the smallest genuinely useful first version
pgvector	An add-on that lets PostgreSQL store embeddings, so no separate AI database is needed
PO / POD	Purchase order to a vendor / proof of delivery signed by the vessel
RAG	AI looks up company records first, then answers from them
RBAC	Role-based access: each user sees only what their role allows
RFQ	Request for quotation sent to vendors
Sprint	A fixed two-week block of work ending in something usable
State machine	The fixed sequence of statuses an order moves through

Term	Plain meaning
Tool calling	How an AI agent uses the system's own functions, such as searching vendors
Webhook	An instant notification from one system to another, e.g. new WhatsApp message